FUNGI AND ITS ECONOMIC IMPORTANCE

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📝🍄🍞 Part 1: In-Chapter Point-Wise Notes

🚫🌿 Why Fungi are NOT Plants (General Features)

- **Structure:** Fungi have **no chlorophyll**, and their body is not differentiated into parts like a root, stem, leaves, or flowers.
- **Cellular Nature:** Compared to bacteria, fungi are more highly evolved. They are **eukaryotes** (possessing a true nucleus) and are generally **multicellular**.
- **Classification Fact:** According to the Five Kingdom classification, fungi are officially **not plants**. The reason is that they lack chlorophyll entirely, meaning their mode of nutrition is strictly **heterotrophic** (instead of autotrophic).

🍞 10.1 Fungi – *A General Study (Rhizopus* / Bread Mould)

- **Common Sightings:** The most common fungi are **moulds** (growing on food, bread, fruits, leather, and tree bark) and **mushrooms/toadstools** (seen in the warm, rainy season).
    - *Note:* Most mushrooms are **poisonous**, but some are **edible**.
- **Bread Mould (*Rhizopus*):** Grows on bread, fruits, animal dung, and leather in warm and humid climates.
- **Body Structure:**
    - **Hyphae:** The body consists of a network of transparent, thread-like structures called hyphae.
    - **Mycelium:** The entire mass (or group) of these threads tangled together is called a **mycelium**.
- **Nutrition:** The penetrating hyphae secrete certain enzymes that digest the food directly in the bread substratum and then absorb it. This outside-the-body process is called **extracellular digestion**.

- **Reproduction:** Occurs both asexually and sexually.
    - **Asexual Reproduction:** Starts within a few days of mycelium growth. Some hyphae become erect (called **sporangiophores**) and swell up at their tips.
    - This swollen tip is the **sporangium**, which turns dark as the spores inside it mature.
    - The sporangium eventually bursts to liberate the spores, which are scattered by the wind. If they fall on a suitable substratum with warmth and moisture, they germinate and form a completely new mycelium.

🔵 **Yeast (*Saccharomyces*) - The Unicellular Exception**

- **Nature:** Yeasts are highly unique because they are strictly **unicellular** (one-celled fungi). They occur either singly or in the form of budding chains.
- **Occurrence:** Found freely in the atmosphere and readily grow in all kinds of sugary solutions and plant exudations (like grapes, nectar of flowers, and sugar cane juice).
- **Structure:** The yeast cell is ovoid in shape. It features a distinct cell wall, a true nucleus, fat vacuoles, glycogen granules, and one or more large vacuoles in the cytoplasm.
- **Nutrition:** They can directly absorb simple sugar (glucose). However, complex cane sugar or sucrose must first be broken down into simple sugar by specific enzymes before absorption.
- **Respiration (Strictly Anaerobic):** Yeast respires in the complete absence of air (without using oxygen).
    - *Biochemical Pathway:* Glucose is broken down into **Pyruvic acid**, which is then converted into **Ethanol** + CO_2.
    - *Chemical Equation:* $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + 2ATP$ (Glucose → Ethanol).

- **Economic Importance:** The greatest importance of yeast to mankind is its property of **alcoholic fermentation**.
    - *Commercial Use:* The alcohol (ethanol) produced is of massive commercial importance. When produced for beverages, the quality of the alcohol varies depending on the raw material fermented and the specific kind of yeast employed.

🧀 10.2 Useful and Harmful Fungi

- **A. Useful Fungi:**
    - Several species of *Penicillium* are utilized in processing **flavoured cheeses**.
    - ***Penicillium chrysogenum*** is the most notable species worldwide, as it is the primary source of the very first wonder drug, **penicillin** (discovered by Alexander Fleming).
- **B. Harmful Fungi (Blue Moulds):**
    - Most fungi are actually harmful. They spoil food and cause severe damage to forest trees.
    - Common moulds like ***Penicillium*** and ***Aspergillus*** frequently grow on oranges, lemons, and other citrus fruits, rotting them completely.
    - Some fungi cause irritating skin diseases in humans, such as **ringworm**.
    - *Crucial Warning:* If you see a greenish fungus growing on the outer skin of an overripe/rotting orange, it is indeed a species of *Penicillium*. **However**, it certainly cannot be consumed or used directly as the medicine "penicillin"!

📝 Part 1: In-Chapter Point-Wise Notes (Continued) 🍄

🍷 10.4 Fungi - Role in Industry

10.4.1 Wine - An alcoholic beverage

- **Process:** Wine and other alcoholic beverages (like beer) are the products of **fermentation**.
- **Fermentation Definition:** A process in which microorganisms (yeast and bacteria) break down carbohydrates into simpler products (ethanol or lactic acid) strictly in the **absence of oxygen**.
- **The Reaction:** Grapes (for wine) or barley maltose (for beer) are crushed and fermented in large vessels. The yeast employed is ***Saccharomyces cerevisiae***.
    - *Equation:* $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ (Glucose → Ethanol).
- **The Killing Point:** Ethanol is produced at a concentration of about **12%**. At this exact point, this concentration becomes toxic and **starts killing the yeast cells**. What is left behind is wine.
- **Spirits:** Hard liquors like whisky and gin, which have a much higher alcohol content, are obtained by the **distillation** of the fermented mixture.
- **Health Impact:**
    - Wine is quickly absorbed into the body.
    - In very mild quantities, it may act as a stimulant, but in large quantities, it is harmful.
    - Alcoholics may develop severe body ailments, particularly **liver cirrhosis**.

10.4.2 Baking (Bread-making)

- **Process:** A mixture of flour (atta/maida), salt, and water makes the "dough". The starch converts into sugar.
- **Leavening:** When yeast is added, it ferments the sugar and rapidly produces carbon dioxide. This gas causes the dough to "rise" (leavening) to about **three times its original volume**.
- **Baking:** When the risen dough is baked, the trapped gas bubbles expand and escape, giving the bread a light, **"spongy" texture**.

10.4.3 Cheese Production

- Cheese is a valuable food high in fat, protein, calcium, phosphorus, vitamin A, and some vitamin B.
- **The Four Major Steps:**
    1. **Curdling of milk:** Brought about by the addition of lactic acid bacteria (*Lactobacillus*). The solid **curd** produced is separated from the liquid **whey** (the liquid remained after curdling).
    2. **Processing:** The curd is processed to remove moisture. At this early stage, it is called **cottage cheese**.
    3. **Salting:** Further removes moisture and prevents the growth of undesirable microorganisms.
    4. **Ripening:** The curd is kept at suitable temperature and humidity. The microorganisms added along with the lactic acid bacteria in step 1 now impart the **particular flavours** to the cheese.

🍄 10.4.4 Mushroom Cultivation

- **Nature:** Mushrooms come in various shapes and sizes (some lack a stalk). Not all are edible; most are mildly or severely poisonous.
- **Key Edible Species:**
    1. **White button mushroom (*Agaricus bisporous*)** - The most popular and widely cultivated variety in India.
    2. **Paddy straw mushroom (*Volvariella*)**
    3. **Oyster mushroom (*Pleurotus*)**
- **Cultivation Overview:** Done indoors (requires little land). Grown on compost based on agricultural wastes.

The Five Major Steps of Cultivation:

1. **Composting:**

- Prepared by mixing **wheat or paddy straw**, chicken manure, and organic/inorganic fertilizers.
- The compost is kept at about **50°C for around one week**.

2. **Spawning:**
    - "Mushroom seed" (consisting of the mycelium of the selected mushroom type) is introduced directly into the compost. It is allowed to spread for a couple of days.
3. **Casing:**
    - A thin layer of soil is spread completely over the compost.
    - *Importance:* This is a very important step because it **gives support to the mushroom**, **provides humidity** (prevents quick drying of the compost), and **helps regulate temperature**.
4. **Cropping and Harvesting:** Growth occurs in three principal stages:
    - *Stage 1:* Mycelium (a network of fibrous mass) grows within 2 to 6 weeks.
    - *Stage 2:* Tiny pin heads appear.
    - *Stage 3:* The button stage grows bigger, attaining marketable size, and the full-grown mushrooms are taken out.
5. **Preservation:** Mushrooms are highly perishable. Their shelf life is increased by:
    - **Vacuum cooling** (to remove air).
    - Giving gamma radiation & storing at 15°C.
    - **Freeze drying:** Sublimation of ice from frozen mushrooms at a reduced pressure in a solution of citric acid, **ascorbic acid**, and brine.

⚠️ **Precautions for Cultivation:**

1. Arrangement must be made for circulating air around the beds.

2. Temperature must be kept low (20-25°C) to prevent the growth of pests and diseases.

🥗 Nutritive Value of Mushrooms

- They are excellent sources of vitamins and minerals.
- They contain good amounts of **niacin, pantothenic acid, and biotin**. These vitamins are remarkably well-retained during cooking, canning, and dehydration.
- **Composition:** Fresh mushroom contains about 85-95% moisture, 3% protein, 4% carbohydrate, 0.3-0.4% fat, and 1% minerals and vitamins.
- *CAUTION:* Never try eating wild mushrooms from the field, as they may be fatal. Buy only from reliable shops.

🏆 Part 2: End-of-Chapter Master Lists (Division 1)

Focus: Terminology & The Ultimate Example Bank 🌍🍄

📚 1. Combined Terms List (Glossary)

Biological Term 🔑	Exam-Ready Definition 📝
Mycelium	The entire tangled mass or network of thread-like structures (hyphae) that makes up the body of a fungus.
Hyphae	Transparent, thread-like structures that form the body of a fungus and penetrate the substratum to absorb food.
Extracellular Digestion	The process where fungi secrete enzymes to digest food *outside* their body in the substratum before absorbing it.

Biological Term 🔑	Exam-Ready Definition 📝
Sporangiophore	The erect, vertically growing hyphae that swell up at their tips to bear spores during asexual reproduction.
Sporangium	The swollen, dark tip of the sporangiophore that contains and eventually liberates mature spores.
Fermentation	A process in which microorganisms (like yeast) break down carbohydrates into simpler products (ethanol or lactic acid) strictly in the **absence of oxygen**.
Leavening	The process where added yeast ferments sugar in dough, producing trapped CO_2 gas bubbles that cause the dough to "rise" and expand.
Spawning	The step in mushroom cultivation where "mushroom seed" (mycelium) is introduced into the prepared compost.
Casing	The spreading of a thin layer of soil over the compost to provide support, humidity, and temperature regulation for growing mushrooms.
Liver Cirrhosis	A severe body and liver ailment typically developed by alcoholics due to the prolonged, harmful consumption of alcohol.

🌟 2. Taxonomic Master Example List

Category 📌	Key Exam Examples to Memorize 🎯
Common Moulds (Harmful/Spoilage)	*Rhizopus* (Bread mould), *Penicillium* (grows on citrus fruits), *Aspergillus*.
Medicinal Fungi (Useful)	*Penicillium chrysogenum* (Source of the first antibiotic, Penicillin).
Industrial Fungi (Useful)	*Saccharomyces cerevisiae* (Yeast used in baking and brewing).
Cultivated Edible Mushrooms	*Agaricus bisporous* (White button), *Volvariella* (Paddy straw), *Pleurotus* (Oyster).
Pathogenic (Disease-Causing) Fungi	Fungi causing **Ringworm** in humans; fungi damaging forest trees.
Bacteria in Dairy (Bonus cross-over)	*Lactobacillus* (Lactic acid bacteria used for curdling milk in cheese production).

🏆 Part 2: End-of-Chapter Master Lists (Division 2)

Focus: Crucial Exceptions & Function Matrix ⚙️🔍

🚨 3. The "Ultimate Exceptions" List (Examiners Love These!)

- **The Cellular Exception:** Almost all fungi (like moulds and mushrooms) are multicellular, *except* **Yeast (*Saccharomyces*)**, which is strictly **unicellular** (one-celled).
- **The Plant Kingdom Exception:** Even though they grow in soil/substratum, fungi are **not plants** because they completely lack chlorophyll and are strictly heterotrophic.

- **The Medicine Source Exception:** Penicillin, the first wonder drug used to kill bacteria, is NOT produced by a bacterium—it is the product of a **mould (fungus)**!
- **The Mushroom Exception:** While mushrooms look like a potential food source in the wild, **most mushrooms are poisonous and fatal**. Only a specific few (like *Agaricus*) are edible.
- **The Respiration Exception:** While most eukaryotic organisms require oxygen, Yeast respires **strictly anaerobically** (in the complete absence of air) to produce ethanol and CO_2.

🛠️ 4. Master Function & Location Matrix

Anatomical Structure / Step ⚙️	**Associated Organism / Process** 📍	**Primary Function / Fact** 🎯
Hyphae	*Rhizopus* (Bread Mould)	Penetrate the bread/substratum to secrete enzymes and absorb digested food.
Spores	*Rhizopus* / Mushrooms	Scattered by the wind to germinate and form a new mycelium when warmth and moisture are suitable.
CO_2 **Bubbles**	Yeast in Baking	Expand during the baking process to give bread its light, **"spongy" texture**.
Ethanol (12% Concentration)	Yeast in Brewing	At 12% concentration, the alcohol becomes toxic and **starts killing the**

Anatomical Structure / Step ⚙️	Associated Organism / Process 📍	Primary Function / Fact 🎯
		yeast cells, halting the fermentation process.
Curdling (Step 1)	Cheese Production	*Lactobacillus* bacteria is added to milk to separate the solid **curd** from the liquid **whey**.
Casing Soil	Mushroom Cultivation	Spreads over compost to **give support**, **provide humidity** (prevent quick drying), and **regulate temperature**.
Vacuum Cooling / Freeze Drying	Mushroom Preservation	Highly specific techniques used to drastically increase the shelf life of highly perishable harvested mushrooms.
Niacin, Pantothenic acid, Biotin	Edible Mushrooms	Key retained vitamins that give cultivated mushrooms their excellent nutritive value.